LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192524171
Date	3 October 2026
Day	saturday

SET 9 – ONLINE LEARNING ACTIVITY ANALYSIS

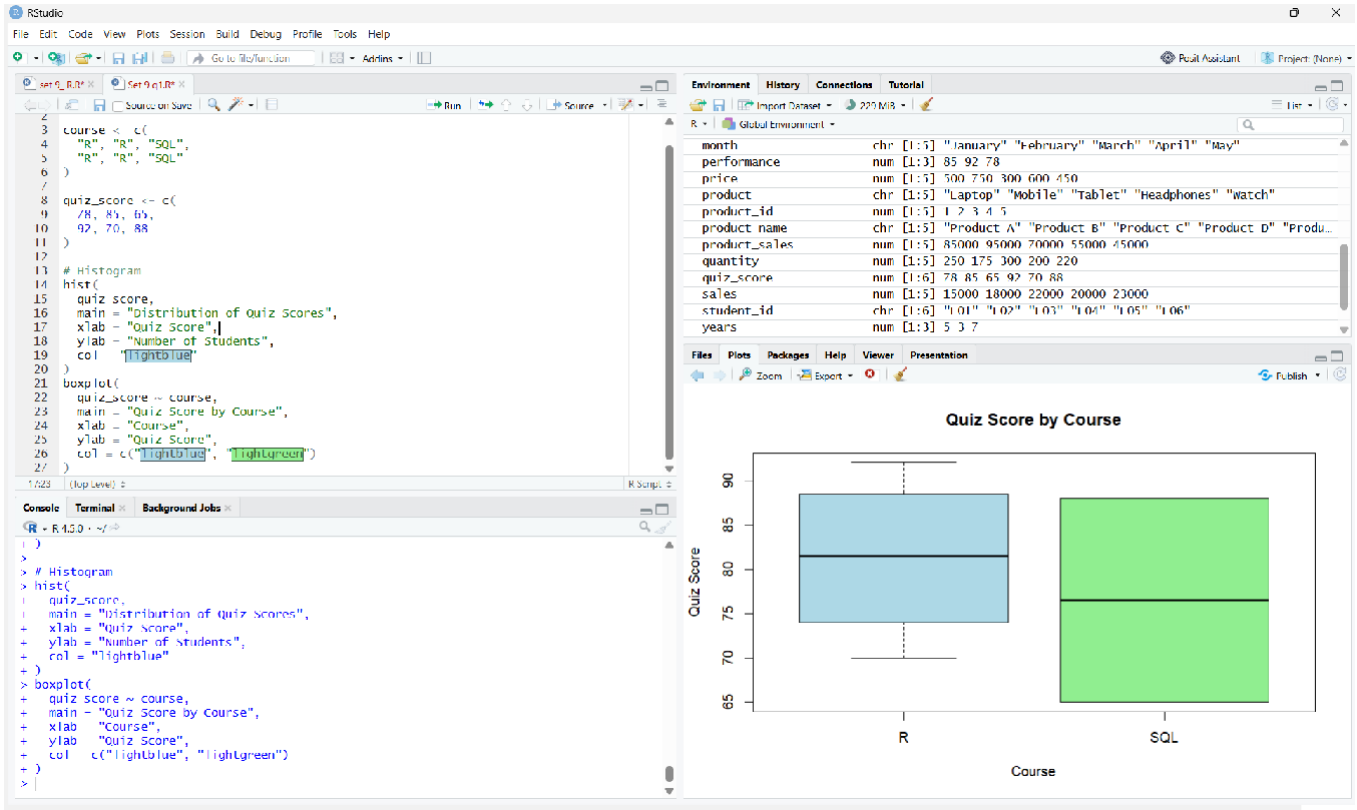
Dataset

Student ID	Gender	Age	Course	Study Time	Videos Watched	Quiz Score	Login Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

PROGRAM 1: Histogram and Boxplot

To analyze student quiz performance using a histogram and boxplot by course and interpret the distribution of quiz scores..

```
boxplot(
quiz_score ~ course,
main = "Quiz Score by Course",
xlab = "Course",
ylab = "Quiz Score",
col = c("lightblue", "lightgreen")
```



Output 1: Daily website page views

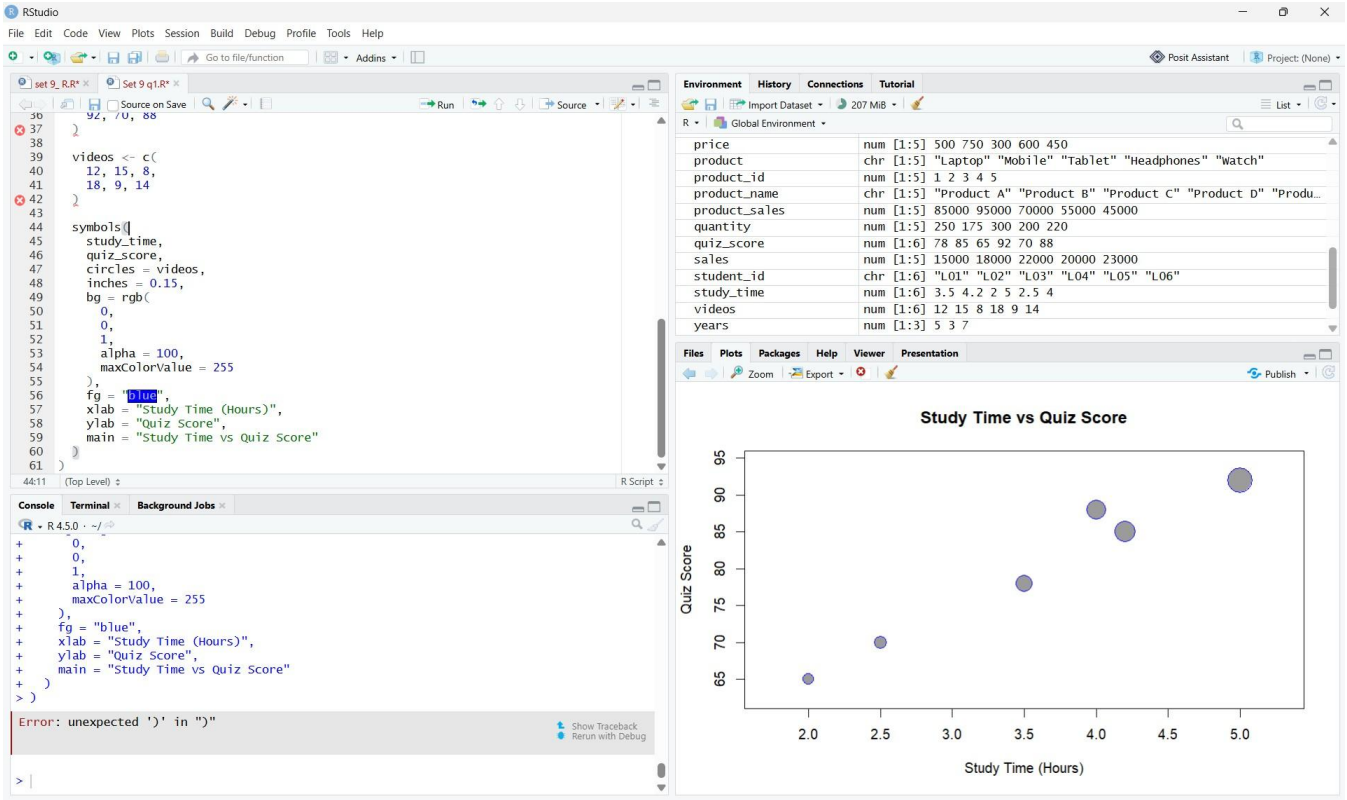
Observation: Quiz scores range from **65 to 92**. The R course contains more students and includes both high and moderate scores. The SQL course has scores of 65 and 88.

PROGRAM 2: Scatter Plot – Study Time vs Quiz Score

To analyze the relationship between study time and quiz score while representing the number of videos watched using bubble size.

```
study_time <- c(
  3.5, 4.2, 2.0,
  5.0, 2.5, 4.0
)
quiz_score <- c(
  78, 85, 65,
  92, 70, 88
)
videos <- c(
  12, 15, 8,
  18, 9, 14
)
symbols(
  study_time,
```

```
  quiz_score,
  circles = videos,
  inches = 0.15,
  bg = rgb(
    0,
    0,
    1,
    alpha = 100,
    maxColorValue = 255
  ),
  fg = "blue",
  xlab = "Study Time (Hours)",
  ylab = "Quiz Score",
  main = "Study Time vs Quiz Score"
)
```



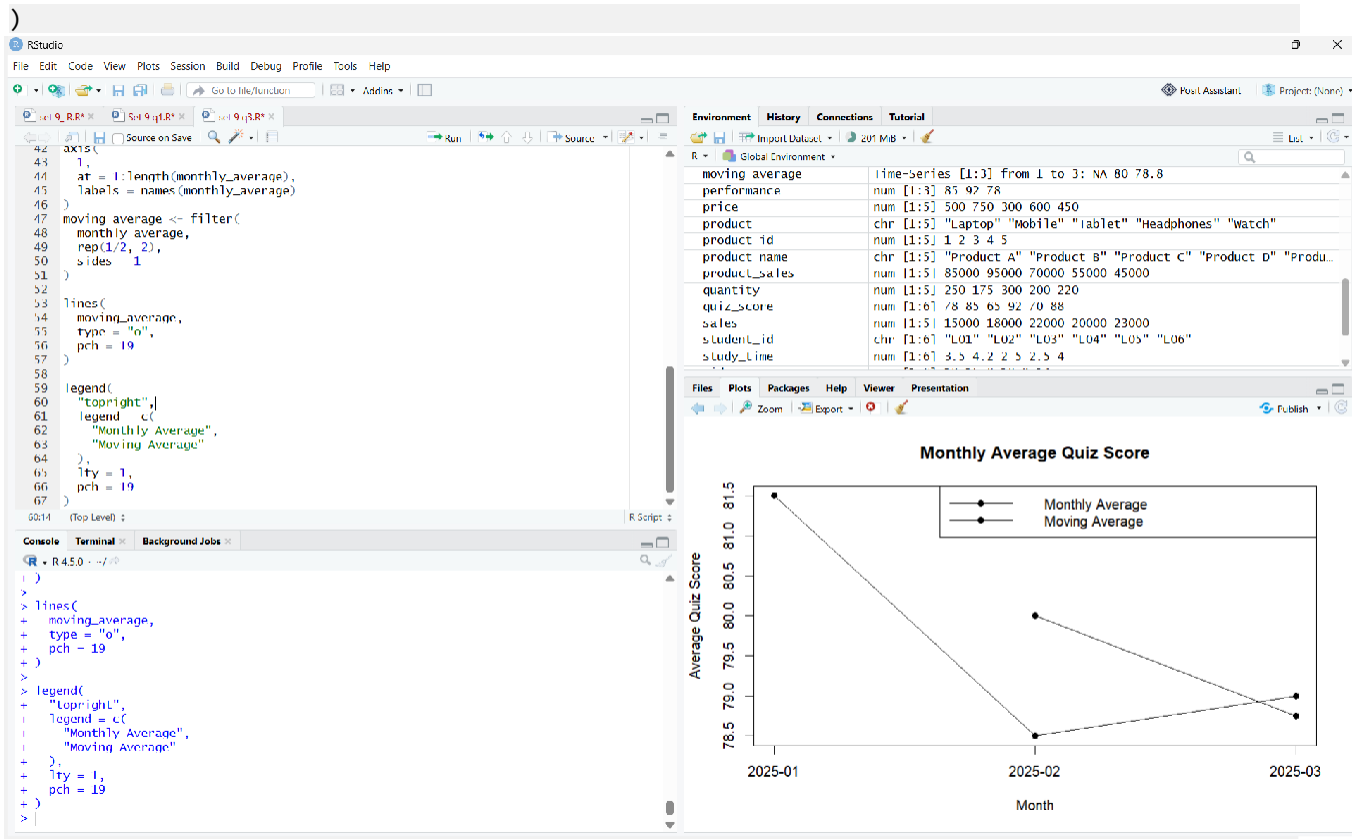
Output 2: click through rate by day

Observation: The data shows a general pattern where students with higher study time tend to have higher quiz scores. Students who watched more videos are represented by larger bubbles.

PROGRAM 3: Monthly Average Quiz Score and Moving Average

To convert login dates into Date format, calculate the average quiz score for each month, and visualize the trend using a line chart with moving average smoothing..

```
moving_average <- filter(
  monthly_average,
  rep(1/2, 2),
  sides = 1
)
lines(
  moving_average,
  type = "o",
  pch = 19
)
legend(
  "topright",
  legend = c(
    "Monthly Average",
    "Moving Average"
  ),
  lty = 1,
  pch = 19
```



Output 3: user interactions

Observation: The average quiz score decreased from **81.5 in January to 78.5 in February**, and then increased slightly to **79.0 in March**. The moving average provides a smoother view of this trend.

PROGRAM 4: Tableau Mini Interactive Dashboard

To create an interactive Tableau dashboard showing average quiz score by course and the relationship between study time and quiz score, with Gender and Course filters

```
library(shiny)

# Dataset

student_id <- c(

  "L01", "L02", "L03",

  "L04", "L05", "L06"

)

gender <- c(

  "Male", "Female", "Male",

  "Female", "Male", "Female"

)

age <- c(
```

```
  20, 22, 19,
  21, 23, 20
)
course <- c(
  "R", "R", "SQL",
  "R", "R", "SQL"
)
study_time <- c(
  3.5, 4.2, 2.0,
  5.0, 2.5, 4.0
)
videos_watched <- c(
  12, 15, 8,
  18, 9, 14
)
quiz_score <- c(
  78, 85, 65,
  92, 70, 88
)
# User Interface
ui <- fluidPage(
  titlePanel(
    "Online Learning Activity Dashboard"
  ),
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "gender",
        "Select Gender:",
        choices = c(
          "All",
          "Male",
          "Female"
        )
      ),

      selectInput(
        "course",
```

```
        "Select Course:",
        choices = c(
          "All",
          "R",
          "SQL"
        )
      )
    ),
    mainPanel(
      tabsetPanel(
        tabPanel(
          "Average Quiz Score",
          plotOutput("barplot")
        ),

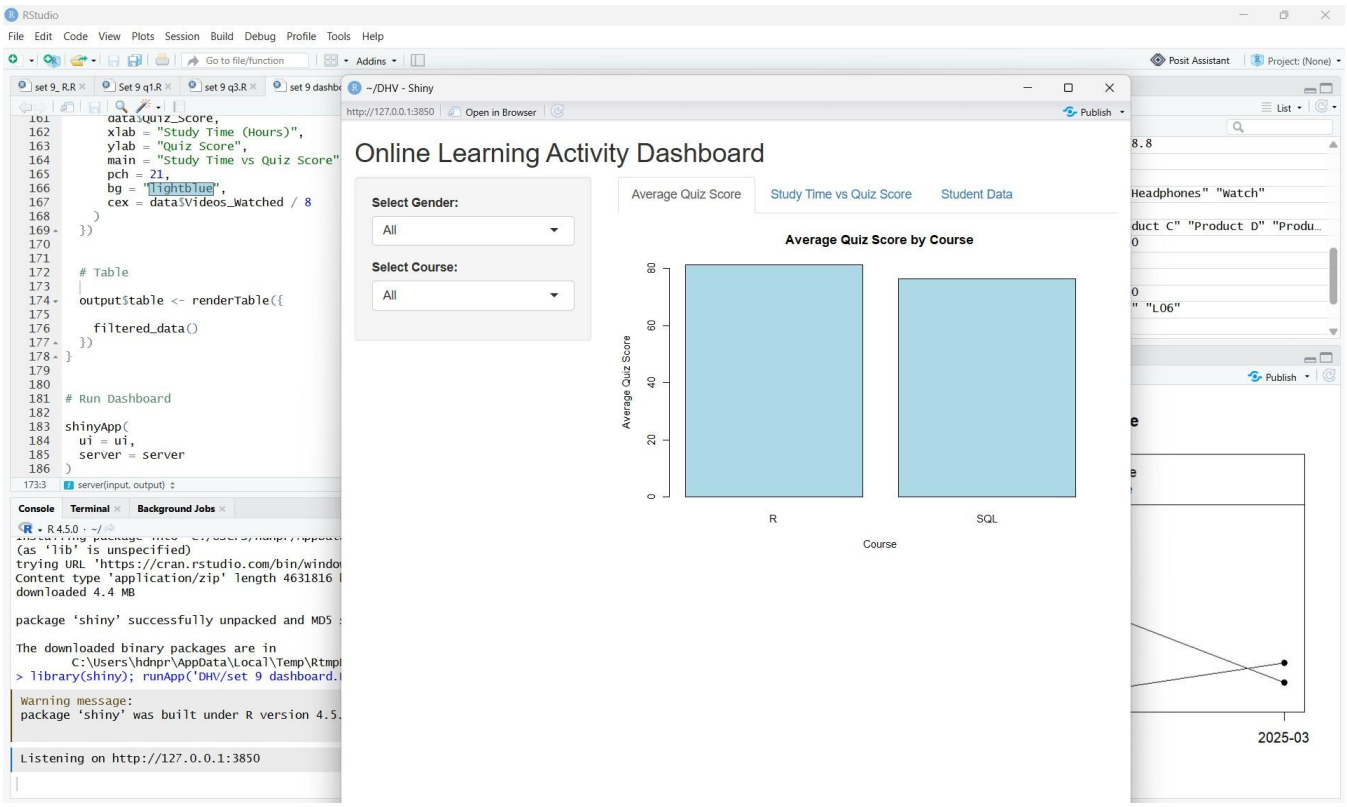
        tabPanel(
          "Study Time vs Quiz Score",
          plotOutput("scatterplot")
        ),

        tabPanel(
          "Student Data",
          tableOutput("table")
        )
      )
    )
  )
)
# Server
server <- function(input, output)
  filtered_data <- reactive({
    data <- data.frame(
      Student_ID = student_id,
      Gender = gender,
      Age = age,
      Course = course,
      Study_Time = study_time,
      Videos_Watched = videos_watched,
```

```
    Quiz_Score = quiz_score
  )
  if (input$gender != "All") {
    data <- data[
      data$Gender == input$gender,
    ]
  }
  if (input$course != "All") {
    data <- data[
      data$Course == input$course,
    ]
  }

  data
})
# Bar Chart
output$barplot <- renderPlot({
  data <- filtered_data()
  average_score <- aggregate(
    Quiz_Score ~ Course,
    data = data,
    FUN = mean
  )
  barplot(
    average_score$Quiz_Score,
    names.arg = average_score$Course,
    main = "Average Quiz Score by Course",
    xlab = "Course",
    ylab = "Average Quiz Score",
    col = "lightblue"
  )
})
# Scatter Plot
output$scatterplot <- renderPlot({
  data <- filtered_data()
  plot(
    data$Study_Time,
    data$Quiz_Score,
```

```
      xlab = "Study Time (Hours)",
      ylab = "Quiz Score",
      main = "Study Time vs Quiz Score",
      pch = 21,
      bg = "lightblue",
      cex = data$Videos_Watched / 8
    )
  })
  # Table
  output$table <- renderTable({
    filtered_data()
  })
}
# Run Dashboard
shinyApp(
  ui = ui,
  server = server
)
)
```



Output 4: website analytics dashboard

Observation: The dashboard allows users to compare average quiz scores between R and SQL courses and explore the relationship between study time and quiz score. Gender and course filters allow the data to be examined interactively.

Set 9 Procedure

- Import or enter the Online_Learning_Activity.csv dataset into R.
- Analyze Quiz_Score using a histogram and boxplot by course.
- Create a bubble scatter plot to study the relationship between study time and quiz score, using videos watched as bubble size.
- Convert Login_Date into Date format and calculate monthly average quiz scores.
- Create a line chart with a 2-month moving average to identify the performance trend.
- Import the CSV file into Tableau.
- Create an average quiz score bar chart and a study-time versus quiz-score scatter plot.
- Add Gender and Course filters and combine the visualizations into an interactive dashboard..

Set 9 Result

The online learning activity data was successfully analyzed using histograms, boxplots, bubble scatter plots, and line charts. Monthly quiz performance and learning patterns were identified, and an interactive Tableau dashboard was created for exploring course, gender, study time, and quiz performance.